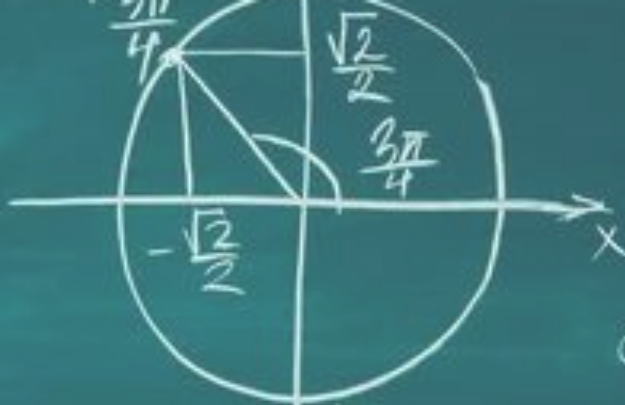
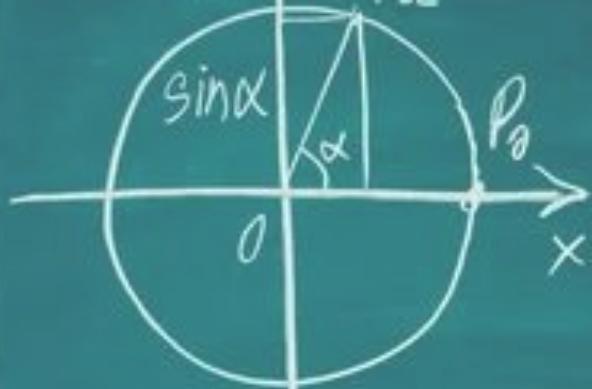




$$\operatorname{tg} \frac{3\pi}{4} = -1$$

$$\sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}, \quad \cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}, \quad \operatorname{tg} \frac{3\pi}{4} = -1$$

$$D(\sin) = D(\cos) = 0, \quad E(\sin) = E(\cos) = [1; 1]$$

First year Aerospace Engineering

Mathematics Recap - Vectors

Marcos Carrera Valle

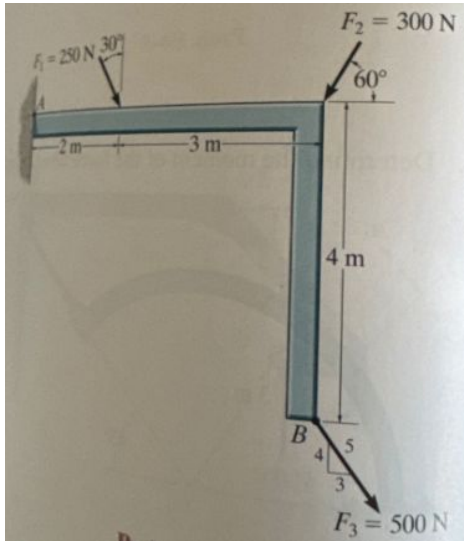
Magnitude and direction

Find the magnitude and direction of the following vectors:

- 1) $v=(3, 4)$ (angle θ with the x-axis)
- 2) $u=(1,-2,2)$ (All 3 angles (α,β,γ) with respect to x-, y-, z-axes)
- 3) $w=(-3,4,-12)$ (unit vector in the same direction)
- 4) $p=(1,-1,2,2)$ (unit vector in the same direction)

Vector decomposition

- a) Determine the moment of each of the three forces about point A
- b) Determine the moment of each of the three forces about point B



Dot product

$$u=(1,-2,2)$$

$$w=(-3,4,-12)$$

- a) Find the angle between u and w .
- b) Find the component of w parallel to u .

Parametric equations, tangents

At $t = 0$ [min], the position of the plane is $(0, 5000)$. Suppose the 2D velocity vector of a plane is given by $V(t) = (3t+50, 0.1t^2)$.

- a) What is the acceleration vector $a(t)$? What about the position vector $r(t)$?
- b) Which force is larger, weight or lift? Which force is larger, drag or thrust?
- c) What is the magnitude of the acceleration of the plane when it passes the point $(1711.5, 5308.7)$?
- d) Give me the vector equation tangent to the position curve at this moment.

Centroid

The following lines form a quadrilateral:

Line 1: $\mathbf{r}_1 = (0, 0) + \lambda (4, 1)$, Line 2: $y = 15 - 3x$

Line 3: $\mathbf{r}_3 = (4, 4) + \lambda (-3, -1)$, Line 4: $\mathbf{r}_4 = (0, 3) + \lambda (2, -2)$

- a) Give me the cartesian form of line 1 and line 3
- b) Give me the vector form of line 2
- c) Give me the centroid of the quadrilateral

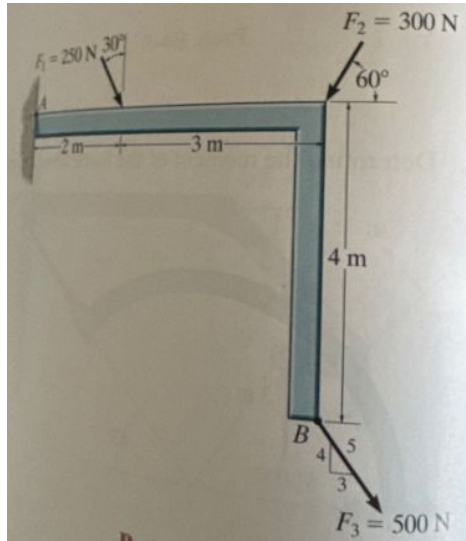
Magnitude and direction

Find the magnitude and direction of the following vectors:

- 1) $v=(3, 4)$ (angle θ with the x-axis)
Magnitude = 5, $\theta = 53.13^\circ$
- 2) $u=(1,-2,2)$ (All 3 angles (α,β,γ) with respect to x-, y-, z-axes)
Magnitude = 3, $\alpha = 70.53^\circ$, $\beta = 131.81^\circ$, $\gamma = 48.19^\circ$
- 3) $w=(-3,4,-12)$ (unit vector in the same direction)
Magnitude = 13, $w_{\text{unit}}=(-3/13, 4/13, -12/13)$
- 4) $p=(1,-1,2,2)$ (unit vector in the same direction)
Magnitude = $\sqrt{10}$, $p_{\text{unit}}=(1/\sqrt{10}, -1/\sqrt{10}, 2/\sqrt{10}, 2/\sqrt{10})$

Vector decomposition

- a) 1983 Nm
- b) -750 Nm



Dot product

$$u=(1,-2,2)$$

$$w=(-3,4,-12)$$

- a) Find the angle between u and w . 153.8°
- b) Find the component of w parallel to u . $(-35/39, 70/79, -70/79)$

Parametric equations, tangents

At $t = 0$ [min], the position of the plane is $(0, 5000)$. Suppose the 2D velocity vector of a plane is given by $V(t) = (3t+50, 0.1t^2)$.

- a) What is the acceleration vector $a(t)$? What about the position vector $r(t)$?
 $a(t) = (3, 0.2t)$, $r(t) = (3/2t^2+50t, 1/30t^3+5000)$
- b) Which force is larger, weight or lift? Which force is larger, drag or thrust?
Lift is larger, thrust is larger (as accelerations are positive at any give moment t)
- c) What is the magnitude of the acceleration of the plane when it passes the point $(1711.5, 5308.7)$?
 $a = 5.16$
- d) Give me the vector equation tangent to the position curve at this moment.
 $r_{\text{tangent}}(t) = (1711.5, 5308.7) + k(113, 44.1)$

Centroid

The following lines form a quadrilateral:

Line 1: $r_1 = (0, 0) + \lambda (4, 1)$, Line 2: $y = 15 - 3x$

Line 3: $r_3 = (4, 4) + \lambda (-3, -1)$, Line 4: $r_4 = (0, 3) + \lambda (2, -2)$

- a) Give me the cartesian form of line 1 and line 3

$$y_1 = (1/4)x_1, y_3 = (x_3 + 8)/3$$

- b) Give me the vector form of line 2

$$r_2 = (5, 0) + \lambda (-1, 3)$$

- c) Give me the centroid of the quadrilateral

$$T_c = (2.74, 2.1)$$